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Subject : Lab 3

Department : BSCS (1st semester)

Section : C

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Notes

Task 1: Input Three Variables

Question: Write a single cin statement to read age, pay, and section.

Solution:

```
#include <iostream>
using namespace std;

int main()
{
    int age;
    double pay;
    char section;

    cout << "Enter age, pay, and section:";
    cin >> age >> pay >> section;

    return 0;
}
```

Notes

Task 2: Evaluate the Expressions

Find the values of the following expressions:

No.	Expression	Answer
1.	28 / 4 - 2	5
2.	6 + 12 * 2 - 8	22
3.	4 + 8 * 2	20
4.	6 + 17 % 3 - 2	6
5.	2 + 22 * (9 - 7)	46
6.	(8 + 7) * 2	30
7.	(16 + 7) % 2 - 1	0
8.	12 / (10 - 6)	3
9.	(19 - 3) * (2 + 2) / 4	16

Remember:

In C++, multiplication, division, and modulus (%) are performed before addition and subtraction. Parentheses are evaluated first.

Rapida Notes

Task 3 : Automatic Data Type Conversion

Given:

```
int units;
float mass;
double weight;

weight = mass * units;
```

Answer: Option C

Explanation:

1. units is converted to float during multiplication.

2. mass * units produces a floating-point result.

3. The result is converted to double when assigned to weight.

Register Notes

Task 4: Type Conversion

Given:

```
int a, b = 2;
float c = 4.2;
a = b * c;
```

Answer: 8

Explanation:

1. b = 2
2. c = 4.2
3. Multiplication: 2 * 4.2 = 8.4
4. Since a is an integer, the fractional part is discarded.

Therefore, a = 8.

Notes

Task 5: Type Casting

Question:

Rewrite the following statement to avoid integer division.

Original statement:

```
unitsEach = qty / salesReps;
```

Solution:

```
unitsEach = static_cast<double>(qty) /
                                salesReps;
```

Explanation:

Converting qty to double before division allows the result to contain a fractional part.

For example, 7/2 using integer division gives 3, but 7.0/2 gives 3.5.

Revatla Notes

Task 6: Math Tutor

Question:

Write a program that displays two random numbers for addition and reveals the answer when the student presses Enter.

Solution:

```
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;

int main()
{
    srand(static_cast<unsigned int>(time(0)));
    int num1 = rand() % 900 + 100;
    int num2 = rand() % 900 + 100;
    cout << "Solve the following addition problem:\n\n";
    cout << "   " << num1 << endl;
    cout << "+  " << num2 << endl;
    cout << "-----\n";

    cout << "\nPress Enter to see the answer.";
    cin.get();
    cout << "\nAnswer:" << num1 + num2 << endl;
    return 0;
}
```

Explanation:

1. srand() initializes the random number generator.
2. rand() generates random numbers.
3. Both numbers range from 100 to 999.
4. cin.get() waits for the student to press Enter.
5. The program displays the correct sum.

Rabbit Notes

LAB 03 — HOME TASKS

Home Task 1: Find and Correct the Errors

Program 1: Addition of Two Numbers

Corrected program:

```
#include <iostream>
using namespace std;

int main()
{
    double number1, number2, sum;

    cout << "Enter a number: ";
    cin >> number1;

    cout << "Enter another number: ";
    cin >> number2;

    sum = number1 + number2;

    cout << "The sum of the two numbers is "
         << sum << endl;

    return 0;
}
```

Pakistan Notes

```
else
{
    quotient = static_cast<double>(number1)/
                                    number2;
    cout << "Quotient = " << quotient << endl;
}

    return 0;
}
```

Errors corrected:

1. Added the required header file.
2. Corrected the main() function.
3. Corrected input and output statements.
4. Used static_cast<double>() to avoid integer division.
5. Added a check for division by zero.
6. Added missing semicolons.
7. Added return 0;.

Rajdeep Notes

Home Task 2: Calculate the Average of Five Numbers

Question:

Calculate the average of the numbers 28, 32, 37, 24, and 33.

Solution:

```
#include <iostream>
using namespace std;

int main()
{
    int num1 = 28;
    int num2 = 32;
    int num3 = 37;
    int num4 = 24;
    int num5 = 33;

    int sum;
    double average;

    sum = num1 + num2 + num3 + num4 + num5;
```

Notes

```
    average = static_cast<double>(sum) / 5;
    cout << "Sum = " << sum << endl;
    cout << "Average = " << average << endl;

    return 0;
}
```

Calculation:

Sum:

$$28 + 32 + 37 + 24 + 33 = 154$$

Average:

$$\text{Average} = \frac{\text{Sum}}{\text{Number of values}}$$

$$\text{Average} = \frac{154}{5} = 30.8$$

Output:

```
Sum = 154
Average = 30.8
```

Regalia Notes